

16x16 2D Systolic Array Hardware Accelerator

Vivado 2025.1 ML Edition: Complete Simulation, Synthesis & Hardware Guide

Target Board: TUL PYNQ-Z2 (Zynq-7000 xc7z020clg400-1)
Baseline Board: Digilent Basys 3 (Artix-7 xc7a35tcpg236-1)

Array Scale: 16x16 Mesh (256 Processing Elements)
Precision: INT8 Inputs (Signed) -> INT16 Accumulation
Core Frequency: 125 MHz (PYNQ-Z2) / 100 MHz (Basys 3)

1. Self-Contained Board Directory Structure

To make file access seamless in Vivado without hunting across scattered folders, every board folder is **100% self-contained** with three dedicated subfolders: **board_implementation/**, **src/**, and **sim/**.

Board Folder	Subfolder	Key Files	Hardware Role & Details
boards/pynq_z2/	board_implementation/	pynq_z2_demo_16x16_top.sv pynq_z2_constraints.xdc	Top hardware demo wrapper with ROM test matrices, LEDs, RGB LEDs, buttons, and 115200 baud UART stream. Master XDC constraints for xc7z020clg400-1.
boards/pynq_z2/	src/	systolic_top.sv, systolic_array.sv, processing_element.sv, skew_buffer.sv, controller.sv, systolic_pkg.sv	Full synthesizable SystemVerilog core modules. Top RTL module is systolic_top.sv .
boards/pynq_z2/	sim/	tb_step6_systolic_16x16.sv tb_matrix_top.sv Matrix*.sv, matrix_if.sv	Contains BOTH the unit testbench (tb_step6) and the enterprise OOP randomized verification suite (tb_matrix_top).
boards/basys3/	board_implementation/	basys3_demo_16x16_top.sv basys3_constraints.xdc seven_segment_ctrl.sv	Basys 3 demo top module driving 4-digit 7-segment display showing 16-bit hex matrix output elements.

2. Design Progression & Why Step 3 is Named 'skew_buffer'

Systolic array matrix multiplication dimensions scale by powers of 2 (2x2, 4x4, 8x8, 16x16). There is no '3x3' matrix test. Instead, **Step 3** represents the development and standalone verification of the **Skew Buffer** component:

Step	Testbench File	What It Verifies
Step 1	tb_step1_pe.sv	Single Processing Element Multiply-Accumulate (MAC) & horizontal/vertical forwarding.
Step 2	tb_step2_systolic_2x2.sv	2x2 (4 PEs) baseline systolic array mesh.
Step 3	tb_step3_skew_buffer.sv	Skew Buffer module alone: row-by-row time staggering (Row 0=0 cycles, Row 1=1 cycle, etc.) into diagonal wave-fronts.
Step 4	tb_step4_systolic_4x4.sv	4x4 (16 PEs) intermediate systolic array mesh.
Step 5	tb_step5_systolic_8x8.sv	8x8 (64 PEs) intermediate scaling test.
Step 6	tb_step6_systolic_16x16.sv	Full 16x16 (256 PEs) top-level accelerator with Identity, Scaled Identity, and Dense Signed GEMM.

3. Vivado 2025.1 Top Module Selection Guide

In Vivado, you must set the Top Module depending on whether you are running a **Simulation** or generating a **Hardware Bitstream**:

Workflow Task	Top Module Name	File Location in Project	Action in Vivado GUI
16x16 Unit Simulation	tb_step6_systolic_16x16	boards/pynq_z2/sim/tb_step6_systolic_16x16.sv	In Sources -> Simulation Sources , right-click tb_step6_systolic_16x16.sv -> Set as Top . Click <i>Run Simulation</i> -> <i>Run Behavioral Simulation</i> .
OOP Randomized Verification	tb_matrix_top	boards/pynq_z2/sim/tb_matrix_top.sv	In Sources -> Simulation Sources , right-click tb_matrix_top.sv -> Set as Top . Click <i>Run Behavioral Simulation</i> .
PYNQ-Z2 Bitstream (Hardware)	pynq_z2_demo_16x16_top	boards/pynq_z2/board_implementation/pynq_z2_demo_16x16_top.sv	In Sources -> Design Sources , right-click pynq_z2_demo_16x16_top.sv -> Set as Top . Click <i>Generate Bitstream</i> .
Basys 3 Bitstream (Hardware)	basys3_demo_16x16_top	boards/basys3/board_implementation/basys3_demo_16x16_top.sv	In Sources -> Design Sources , right-click basys3_demo_16x16_top.sv -> Set as Top . Click <i>Generate Bitstream</i> .

4. Conventional Unit TB vs. Enterprise OOP Verification

Conventional Unit TB (tb_step6): Uses procedural initial/task blocks to feed 3 fixed test matrices (Identity, Scaled Identity, Dense GEMM). Best for fast waveform debugging.

Enterprise OOP Verification (tb_matrix_top): Uses class-based architecture modeled after IEEE UVM (Universal Verification Methodology):

- **MatrixTransaction:** Encapsulates matrices A, B, output C, and execution cycles.
- **MatrixGenerator:** Automatically generates 30 randomized stress-test matrices with boundary conditions (-128 to +127, zeros, sparse).
- **MatrixDriver:** Wiggles physical interface pins cycle-by-cycle.
- **MatrixMonitor:** Samples output matrix C upon assertion of the done signal.
- **MatrixScoreboard:** Computes mathematical golden product and verifies all 7,680 elements with 0 mismatches.

5. How to View Output Matrix on PYNQ-Z2 (No 7-Segment Display)

The Basys 3 has an on-board 4-digit 7-segment display, but PYNQ-Z2 does not. Here are the 4 practical ways to view results on PYNQ-Z2:

1. Built-in 115200 Baud UART Serial Monitor (Already Implemented in our RTL!):

- The top module transmits results over PMOD A Pin 1 (Pin **Y18**).
- Plug in a USB-UART adapter or micro-USB to your laptop, open PuTTY or VS Code Serial Monitor at 115200 baud, 8N1. The board prints 16x16 GEMM: PASS and matrix output automatically upon completion.

2. On-Board Diagnostic Status LEDs & RGB LEDs:

- **led[0] (Heartbeat)**: Blinks at ~1.86 Hz proving the 125 MHz clock is running.
- **led[1] (Busy)**: High while the 256 PEs are computing (48 cycles).
- **led[2] (DONE)**: Turns solid bright when matrix C is calculated.
- **led[3] (PASS)**: Onboard hardware comparator confirms all 256 elements match expected golden results.
- **rgbled4**: Blue = IDLE, Amber = COMPUTING, Vibrant Green = 100% VERIFIED PASS.

3. External PMOD 7-Segment / I2C OLED Display:

- Connect a Digilent Pmod SSD (dual 7-segment) to PMOD A, or a \$3 I2C 0.96" OLED screen (SSD1306) to Arduino SCL/SDA pins to display text & values.

4. PYNQ Jupyter Notebook (Embedded Python Overlay):

- PYNQ-Z2 has dual ARM Cortex-A9 running Linux. Through Jupyter Notebook (<http://pynq:9090>), load the bitstream and plot a 2D color heatmap of matrix C using Matplotlib!